

VISUALIZING THE WEB SEARCH RESULTS WITH *WEB SEARCH VISUALIZATION USING SCATTER PLOT*

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ABSTRACT

We live in 21st century, a time when the speed of flow of information is the indicator of the quality of life. Therefore, it is rightly called the century of information. We need information in our everyday life, but we need to get it in bulk, with ease and with assured quality. The best and easiest way to get it is through the World Wide Web. Although there are a good number of search engines available in web, the most commonly used ones show the search result in a list based representation. This list based representation is not very user friendly for the users all the time. Sometimes they don't get their desired results easily and there is not enough scope for the user to compare documents, find relevant documents and filter according to their needs. *Web search visualization using scatter plot* provides the users to search their documents and the results will be visualized in a traditional list based representation and also in a scatter plot according to the documents' last modification date and document type.

Keywords

Web search visualization, Information visualization, Scatter plot

1. INTRODUCTION

With the advent of information technology and increase of information in the current era of information, our social and industrial life is gradually submerging in massive online information. In this vast ocean of information, search engines are essential to help us maneuver this vast availability properly. Search engines are indeed a valuable gift of modern computer science that has changed our way of accessing, managing and using information. Although a lot of efforts have gone by to create

search engines that can index billions of documents and return them in a fraction of seconds [3]; however the interface has not substantially changed from the early days of these web search engines. Still the results are delivered in a list based representation with 10 results per page usually. Users have to shift through a great number of results which are not often relevant to their interests.

A study [8] from 2006 shows that 88% of users will try a new search if they do not find what they seek in the first three pages. That means in a usual search interface the users would review 30 results and then they will start a new search with a different query or they will modify their previous query to get a different set of search results.

To address the shortcomings in the conventional web search interfaces, *Web search visualization using scatter plot* has been developed. It uses the concept of information visualization to visualize two of the search results attributes, the last modification year and the document type. These two attributes are plotted in a 2D scatter plot. User can visualize these two attributes from the scatter plot and also they can see the conventional list based representation of the search results.

The motivation behind this work is to give the user some scope to see the relation between the search results in respect to their last modified year and the document type. They can compare and analyze the data and find the relevant ones from the list based search result. Suppose the user wants some search results which are last modified in the year 2009, then they can see the scatter plot to find which search results are last modified in 2009. After that they can select them or wish to view them from the actual results which are represented in the conventional list based representation. They also can search for a number of document types like pdf, ms office, and html files for a range of years by the information given in the scatter plot.

The remainder of this paper is organized as follows: An overview and taxonomy of the existing web search visualization techniques is provided in Section 2. In Section 3, the details of *Web search*

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visualization using scatter plot system are briefly described. Conclusion and future work are presented in Sections 4 and 5 respectively.

2. RELATED WORKS

In recent years a number of systems have been proposed and developed to visualize the results of web search in different approaches. While it is not feasible to provide a complete analysis of these systems in this paper, a brief overview of some relevant and interesting techniques is provided here. I briefly describe these according to the means of visualization techniques and the interaction techniques used in these systems.

2.1 Visualization techniques used in existing web search systems

xFind [1] and Hotmap [6] provide a simple list based (1D) representation of search results. Wavelens [12] also provides a 1D visualization technique but in a different manner. It provides a focus+context representation of the search results allowing the user to dynamically zoom into the documents surrogates of interest. The proposed system *Web search visualization using scatter plot* uses a simple list based representation like any other conventional web search system and also that of xFind [1] and Hotmap [6].

2D visualization technique was used in xFind [1], WebQuery [4], Hotmap [6] WebSearchViz [10] and Envision [11]. But the proposed system *Web search visualization using scatter plot* uses the idea of showing two attributes of the search result in a scatter plot from xFind [1] and Envision [11].

xFind [1] uses a scatter plot which is a 2D visualization, to allow interactive exploration of search result set. Documents are plotted according to two of their metadata attributes (corresponding to x and y axes). The relevancy of the matching is plotted in y axis where the document size is plotted in the x axis. The most relevant document is shown in the top of the plot. The color of the document determines the document age, yellow ones are the older ones and whiter ones are the recent ones. The axes can be modified as per the user need. Envision [11] uses a highly customizable scatter plot and iconic visualization to represent the many different attributes available as a part of their information retrieval system.

There are a few more types of visualization techniques which are used in different web search systems like 3D visualization technique was used in WebQuery (in one of its example) [4] and Card Visualization [9]. PEx-Web [13] uses multidimensional space for its visualization of retrieved results. Each document is presented as a point in the multidimensional plane here. WebQuery (in one of its example) [4] uses cone tree visualization for visualizing the search results. WebQuery [4] (in one of its example) and Card Visualization [9] use graph visualization as a visualization technique.

2.2 Interaction techniques used in existing web search systems

There are a number of interaction techniques available which are helpful for the users to conduct their search like selecting, focusing, scrolling or navigation, mouse hovering, zooming, filtering etc.

In most of the existing web search visualization system, selection using mouse click is used. It is used in xFind [1], WebQuery [4], Hotmap [6], WebSearchViz [10], Card Visualization [9] and PEx-Web [13]. This interaction technique is helpful for the users to select their desired search results easily with the help of a mouse in a fraction of seconds.

Interaction techniques like focus or overview has its importance to the users. It is used to gain a big picture view of the entire search results or a number of search results. Wavelens [12] and Card Visualization [9] have this feature.

Another important interaction technique for the user is to scroll or navigate through the search result window so that they can view or select or do other kind of interaction on all the search results available. Mostly all the conventional web search system has this interaction technique. Hotmap [6], Wavelens [12], WebQuery [4] and Envision [11] also has this feature available for the users.

Mouse hovering is used for special purposes. The programmer of the system chooses what user can do with mouse hovering like tooltips are shown to the user when the user hover his mouse over a document surrogate in Hotmap [6]. When the mouse hovers into any result in Wavelens [12] it becomes the focal point and gets zoomed. PEx-Web [13] also has hovering options.

Zoom is another important interaction technique. It is used to zoom in on the items of interest or zoom out can be used to see the relation of different items in a fixed visual display. Amongst all these existing systems WebSearchViz [10] has zooming options available for the users.

Filtering is another interaction technique which is very useful for the users. A user can remove uninteresting or unneeded items from visual display using filter. It also allows users to focus on the items which remain on the visual display. WebSearchViz [10] and xFind[1] has filtering option amongst these existing techniques.

3. PROPOSED SYSTEM: WEB SEARCH VISUALIZATION USING SCATTER PLOT

The proposed system: *Web search visualization using a scatter plot* is a web search system that retrieves the search results for a given user query from the Yahoo! Search Boss API [16] and presents these search result in two ways.

Web search visualization using Scatterplot

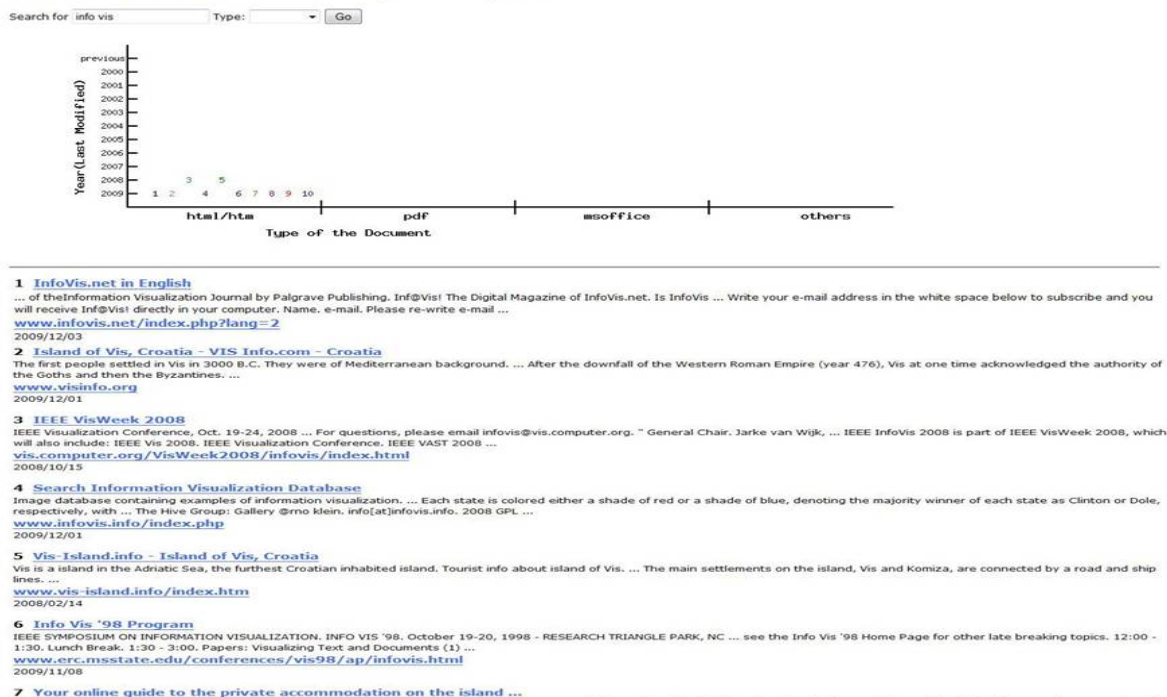


Figure 1: The graphical layout of *Web search visualization using scatter plot*. On the top of the screen the scatter plot showing 10 plotted points for the search result of the query “info vis”. The first 10 results all were basic html/htm files so they all are shown in the area of html/hta division. On the bottom of the screen the actual search results. It’s a list based representation of the results retrieved from Yahoo! Search Boss API.

3.1 System Design

Basic HTML [7] and PHP [14] were used to design this system. The pre defined functions and the response fields of Yahoo! Search Boss API [16] were used to retrieve the search result from Yahoo [15]. This API returns snippets for a query containing the same information that is returned on a search on the Yahoo [15] web site. Apache2Traid [2] was used for server-client communication and as PHP [14] interpreter.

3.2 Detailed description of *Web search visualization using scatter plot*

3.2.1 The Interface

Mainly the users will see a list-based interface with a 2D scatter plot on top of it. There is a text box in the top of the interface where the user can enter their search query and a “Go” button to get the results. There is a drop down menu used also for filtering purpose which will be discussed in Section 3.3.

The scatter plot mainly uses the idea of xFind [1], which could show the date, document relevancy and document size on a scatter plot. Here the scatter plot will show the document type and the date in x and y axes respectively. See Figure 1.

3.2.2 Number of results per page

The list based search representation will show 10 results in each page for the sake of simplicity. The results will be retrieved from Yahoo! Search Boss API [16] each time an user searches using a query. The scatter plot will show the document type and the year of the document for each page. So, for each page there will be 10 points plotted in the scatter plot. The number of results per page are kept down to only 10 so that the user can visualize the points (which are used as numbers from 1-10) in the scatter plot effectively.

3.2.3 Design of the scatter plot

Four kinds of data types are plotted in the x-axis. They are html/htm, pdf, msoffice and others. These are plotted as categorical data in the x-axis.

In the y-axis there are 11 values of year given. There is one value for 2000 to 2009 each and before 2000 all the documents will be plotted in “previous” value. As we only find a small number of search results which has been lastly modified before year the 2000, so for the sake of simplicity all of those years before 2000 are given only the value “previous”. That means if the user get a search result which has lastly modified in the year 1999 will fall into the “previous” value of y-axis and if the user gets any result of the year suppose 1996 then it will also fall in the value “previous” in the y-axis.

In the scatter plot the points will be shown as numbers as it is in the search result. If for a given user query a search result’s ranked position in the list based representation is 7, then in the scatter plot it will also be represented as 7.

3.2.4 Use of color

For showing the font of the ten numbers, ten different colors have been used. Originally the color of these fonts were selected randomly but as it is hard to find out ten different colors which will be perceived by the users easily; so no same type of two colors is used in two fonts which were one after the other. Black, gray, teal, maroon, green, navy, olive, purple, red and blue colors are used for ten numbers in ascending order of magnitude. Although this is not a good idea to select the color of the fonts randomly but for the sake of simplicity it has been done in this version of the proposed system. In the next improved version a thin black border around the fonts will be placed for the users to perceive the fonts easily.

3.3 User Interactions

Not much user interaction technique has been implemented in this version of the system yet. They are referred to the future work. Though some of the user interaction methods used here are briefly described next sub section.

3.3.1 Selection

User can select any specific search result from the list based representation using left mouse click over the title of the result. They can open the URL of the result in a new tab or new window using right mouse click.

3.3.2 Scrolling

User can scroll down and up to the bottom and top of the visual screen to see and interact with all the search result and the scatter plot which is located on the top of the screen.

3.3.3 Filtering Option

There is a filtering option as well in this system. User can filter their search results based on document types (i.e., html, msoffice, pdf). If the user selects any of these document types then only those type of documents will be shown in both the list based representation and the scatter plot. User can filter the type of the document using a drop down menu adjacent to the search box. The drop down menu also used here for the sake of simplicity as the user would have filtered the document type or the easily if there were a interaction method given to them like clicking the different categorical values in the scatter plot and filter from there. It is referred to as future work for this proposed system. Three values are given there; html, pdf and msoffice. User can get html and htm files by choosing “html” option. They will get the pdf files of the search result if “pdf” is selected and user can get MS Word, MS Excel, MS PowerPoint file by choosing “msoffice”. If the users don’t select any of these three options then they will get the normal result set comprising of all the document type. See Figure 2, 3 and 4.

Web search visualization using Scatterplot

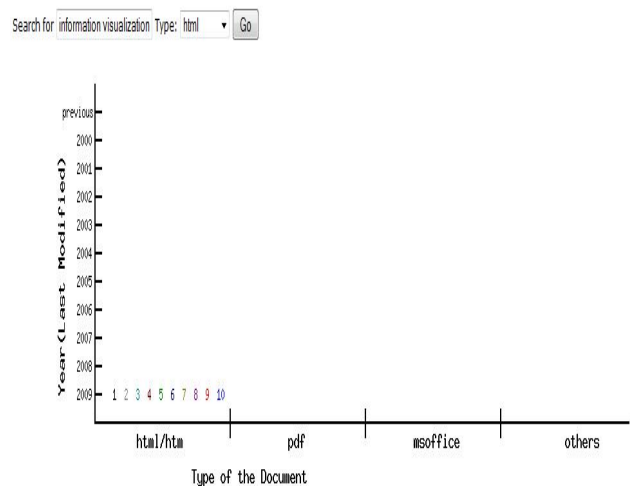


Figure 2: The filtering option in Web search visualization using scatter plot. The user gives the query “information visualization” and selects the document type as “html”. So only the “html” results will be shown in the list based representation and in the scatter plot for the given query.

Web search visualization using Scatterplot

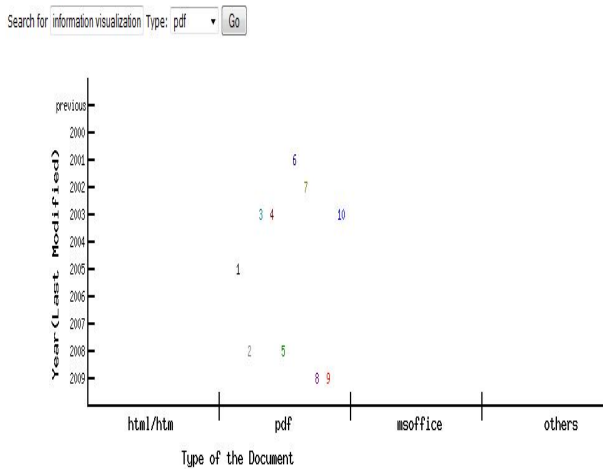


Figure 3: The filtering option in *Web search visualization using scatter plot*. The user gives the query “information visualization” and selects the document type as “pdf”. So only the “pdf” files will be shown in the list based representation and in the scatter plot for the given query.

Web search visualization using Scatterplot

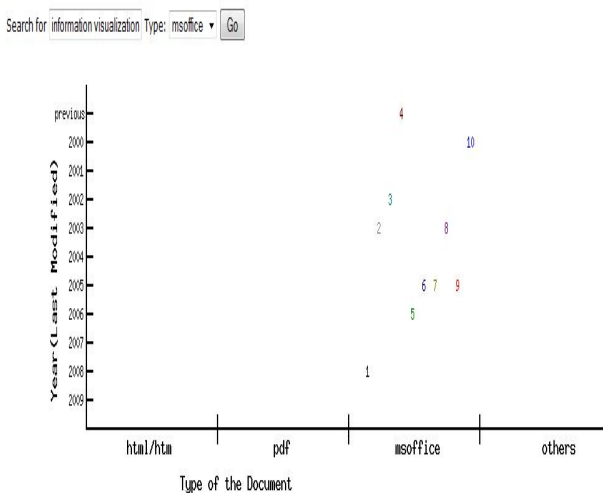


Figure 4: The filtering option in *Web search visualization using scatter plot*. The user gives the query “information visualization” and selects the document type as “msoffice”. So only the “msoffice” files will be shown in the list based representation and in the scatter plot for the given query.

4. CONCLUSION

In this paper, we discussed *Web search visualization using scatter plot*, a visualization technique for visualizing the traditional list based search result in order to give the users more scope to

analyze and manipulate the documents with the help of a scatter plot.

In short, the olden times of machinery in human life is the history of progression, and the *Web search visualization using scatter plot* is a step in advancement of information technology.

5. FUTURE WORK

The scope of user interaction is currently lesser than it is expected in this version of *Web search visualization using scatter plot*. The future work will be to modify this system to incorporate more interaction techniques so that the user can find their document in the easiest possible way.

In the future work there will be an addition of a filter option for the user in respect of the last modified year also. There will be no drop down menus instead the user will be able to click on any year and/or any type of document in the scatter plot to initiate the filtering. The numbers to show the documents in scatter plot will change to small circles as points in the graph. The user will be able to click on any point and the relevant search result will be selected from the list based representation.

And also we are planning to run an user evaluation so that we can compare it with the traditional and popular search engines.

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